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Functional Safety Analysis of the RaSTA Protocol

Understanding how RaSTA enables safe, reliable, and interoperable communication in railway systems.

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SAFE COMMUNICATION | RELIABLE DATA EXCHANGE | INTEROPERABLE SOLUTIONS | BUILT FOR RAILWAYS

RaSTA
Safe Communication for Railways

AGENDA

An overview of the key topics we will cover in this presentation.

1 INTRODUCTION
- Functional Safety
- What is RaSTA?

2 WHY RaSTA?
- Communication Threats
- Why not TCP/IP?

3 RaSTA ARCHITECTURE
- RaSTA Deployment in Railway Systems
- Inside a RaSTA Communication Node

4 HOW RaSTA WORKS
- Connection Establishment
- Safe Data Transmission
- Connection Supervision

5 KEY SAFETY MECHANISMS
- Sequence Numbers
- Integrity Checking
- Heartbeats
- Adaptive Timing
- Retransmission
- Redundancy

6 OPERATIONAL EXAMPLES
- Lost Packet Detection & Retransmission
- Corrupted Message Detection
- Delayed Message Detection
- Heartbeat Failure
- Primary Network Failure

7 REAL-WORLD APPLICATIONS
- Railway Signalling Systems
- ETCS
- Interlockings
- Control Centres

8 ADVANTAGES & LIMITATIONS
- Advantages of RaSTA
- Limitations and Considerations

9 CONCLUSION & DISCUSSION
- Key Takeaways
- Questions & Discussion

> “RaSTA ensures safe, reliable and deterministic communication for the future of railways.

RaSTA is designed to keep railway communication **safe**, **reliable** and **available** – even in the presence of faults.

WHAT IS FUNCTIONAL SAFETY?

Ensuring **safety by design**, even when things go wrong.

RaSTA
Safe Communication for Railways

Functional Safety is part of the overall safety of a system. It refers to a system's ability to **respond to failures** in a safe and controlled way, **reducing risks** to people, the environment, and equipment.

GOAL

Prevent hazardous events or minimize their impact when failures occur.

KEY PRINCIPLES

SAFETY BY DESIGN

Safety is considered in every phase of the system lifecycle, from concept to operation.

IDENTIFY & MANAGE RISKS

Potential hazards are identified and risks are analyzed and reduced to an acceptable level.

DETECT FAILURES

The system can detect faults in time using diagnostics and monitoring.

RESPOND SAFELY

When a fault is detected, the system responds in a safe and controlled manner.

RELIABLE & AVAILABLE

The system continues to operate reliably and is available when needed.

WHY IT MATTERS IN RAILWAYS

- Protects passengers, staff, and the public.
- Ensures safe operation even in the presence of equipment or communication failures.
- Builds trust in railway systems through dependable and predictable behavior.

IN SIMPLE TERMS

Failures can happen (hardware, software, communication, etc.) → Functional Safety ensures these failures are detected in time. → The system reacts in a safe way to prevent accidents or harm. → Result: People, environment and assets are protected.

This is **Functional Safety** — Safety through reliability.

WHAT IS RaSTA?

RaSTA stands for **Rail Safe Transport Application.**

RaSTA

Safe Communication for Railways

RaSTA is a safety-oriented communication protocol used in railway applications. It ensures that data is transmitted **reliably and safely** between devices, **even in the presence of communication failures.**

KEY PURPOSE

SAFETY FIRST

Detects communication errors and prevents them from leading to unsafe situations.

RELIABLE COMMUNICATION

Ensures messages are delivered in the correct order, without loss or duplication.

CONTINUOUS SUPERVISION

Monitors the connection constantly using heartbeats and sequence numbers.

SECURE & DETERMINISTIC

Provides integrity protection and predictable behavior for safety-critical systems.

BUILT FOR RAILWAYS

Designed specifically for railway control and signalling applications with high safety demands.

STANDARDIZED BY

RaSTA is standardized in

DIN VDE V 0831-200

(RaSTA Specification)

It defines the protocol requirements to achieve safe and reliable communication in railway systems.

IN SIMPLE TERMS

RaSTA enables devices in a railway system → to communicate safely and reliably → by detecting errors quickly and ensuring correct data exchange → so that railway operations remain safe, dependable and always under control.

RaSTA = Safe Communication for Safer Railways

WHY RaSTA IS NEEDED

RaSTA ensures safe, reliable and secure communication in railway systems.

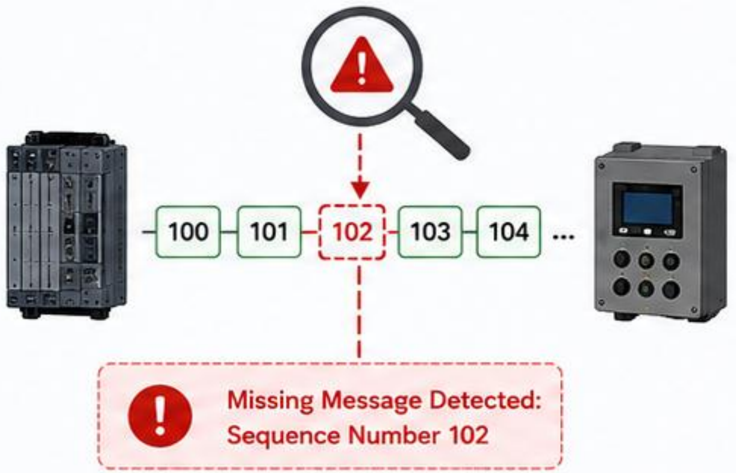
1 Prevents Hazardous Failures

Ensures that communication errors do not lead to accidents or unsafe train operations.

RaSTA keeps communication safe so failures never become accidents.

2 Detects Faults Quickly

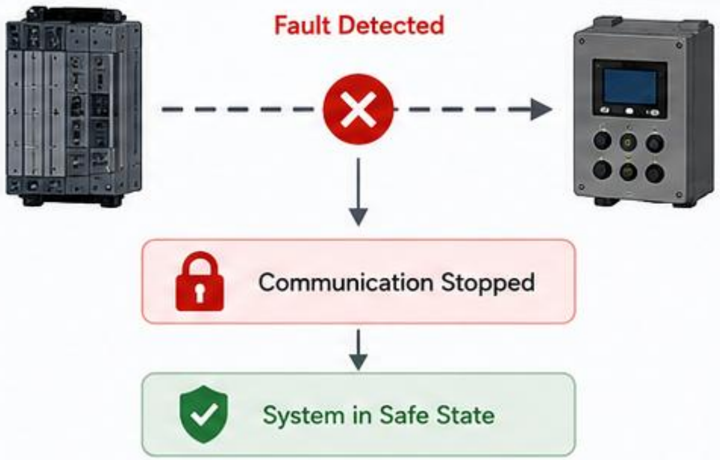
Identifies communication or system errors such as lost, corrupted or out-of-order messages.



RaSTA detects problems **early** before they impact operations.

3 Moves to a Safe State

Stops or disconnects the system when a fault is detected to prevent any unsafe action.



RaSTA ensures the system switches to a **safe state** automatically.

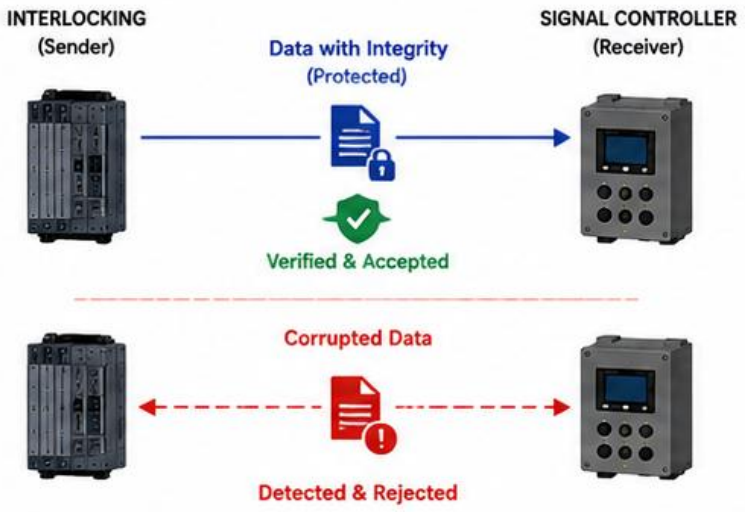
These capabilities make RaSTA essential for **reliable and safe** railway communication.

WHY RaSTA IS NEEDED – NEXT 3

RaSTA ensures safe, reliable and secure communication in railway systems.

4 Ensures Data Integrity

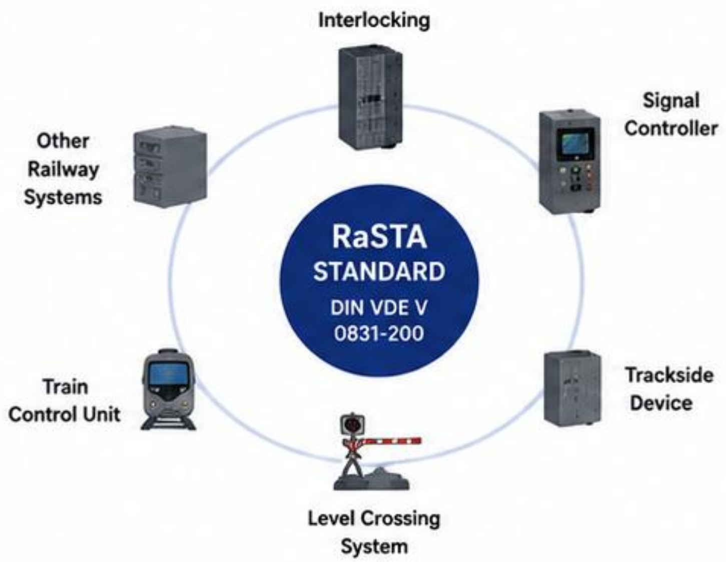
Guarantees that the data received is exactly what was sent, without any alteration or corruption.



RaSTA uses security mechanisms to ensure data is **correct and trustworthy.**

5 Supports Interoperability

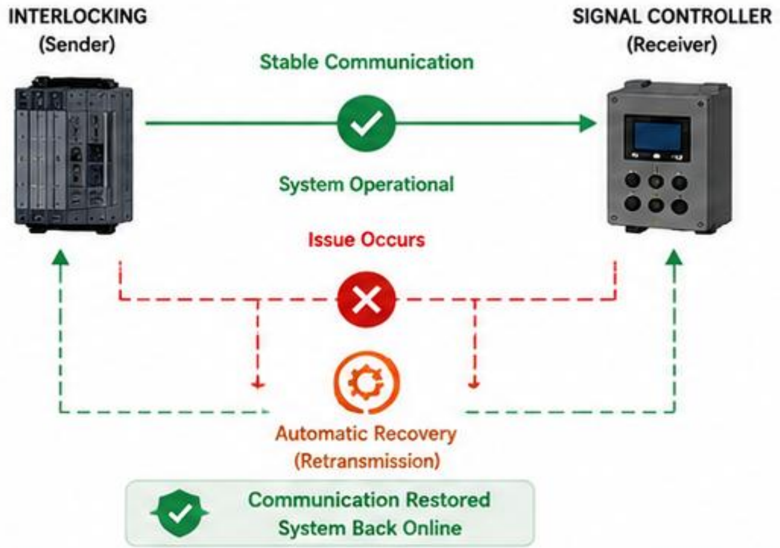
Standardized by DIN VDE V 0831-200, enabling different devices and systems to communicate seamlessly.



RaSTA allows devices from different manufacturers to **work together** reliably.

6 Enhances System Availability

Minimizes downtime by ensuring continuous monitoring and automatic recovery from communication issues.



RaSTA keeps the system **available** and reduces service interruptions.

Together, these capabilities make RaSTA essential for **safe, reliable, secure** and **interoperable** railway communication.

WHY NOT TCP/IP?

TCP provides reliable delivery, but **not functional safety**.

TCP/IP

TCP/IP	VS	RaSTA
Reliable packet delivery	Primary Goal	Safety-critical communication
Detects transmission errors	Error Detection	Detects safety hazards and protocol errors
Retransmits lost packets	Recovery	Retransmits while preserving safety
No replay or sequence safety	Sequence & Replay Protection	Prevents replay, duplication, and reordering
Cannot guarantee fail-safe operation	Failure Handling	Disconnects safely when faults occur

RaSTA

Generic Network Communication

Safe Railway Communication

RaSTA builds on standard networks but adds the safety mechanisms required for railway signalling.

RaSTA ARCHITECTURE

A safe, reliable and independent communication system for railway applications.

RaSTA
Safe Communication for Railways

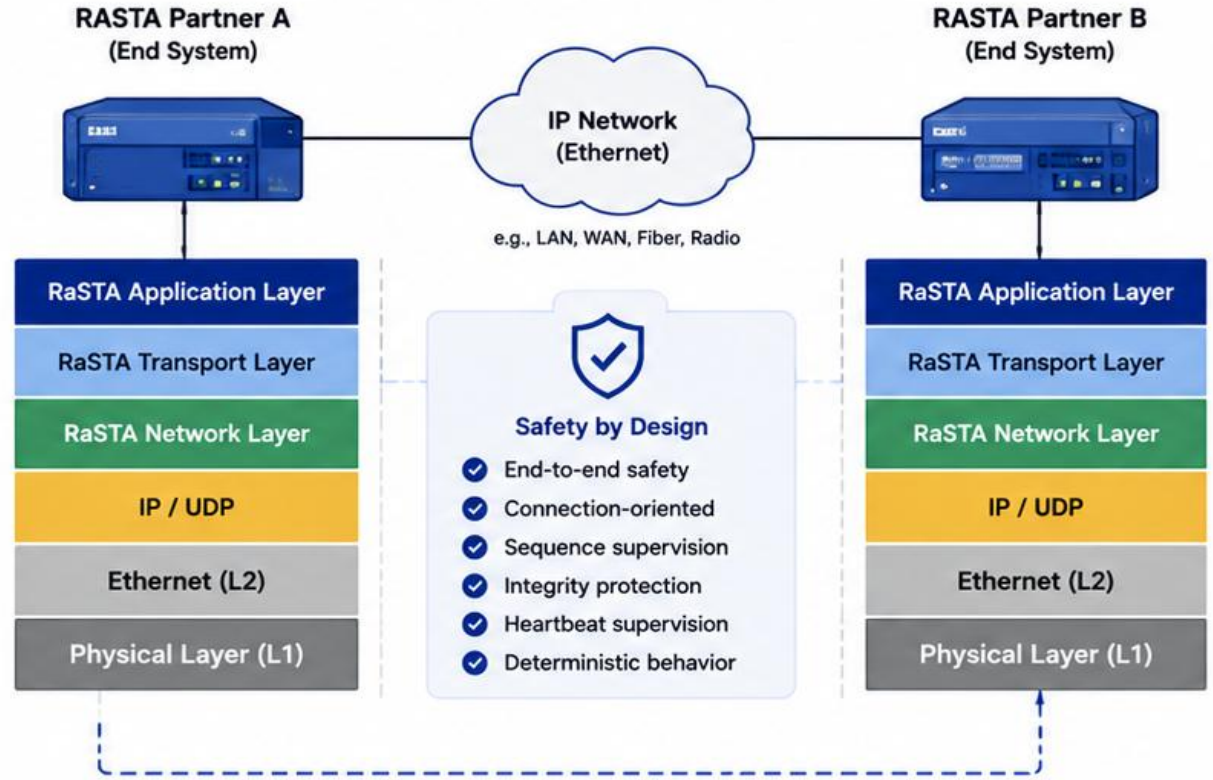
WHAT IS RASTA?

RaSTA (Rail Safe Transport Application) is a safety-oriented communication protocol standardized in DIN VDE V 0831-200.

It runs over standard IP networks but adds its own safety mechanisms.

Designed for safety-critical railway functions where communication failures must be detected and handled safely.

RASTA COMMUNICATION ARCHITECTURE



LAYERS & COMPONENTS

RaSTA Application Layer
Provides services to the application. Defines communication endpoints and data handling.

RaSTA Transport Layer
Manages connections, segmentation, acknowledgements and flow control.

RaSTA Network Layer
Handles addressing, routing information and path management.

IP / UDP
Standard IP network and UDP are used as the transport medium.

Ethernet (L2)
Data link layer for frame delivery on the local network.

Physical Layer (L1)
Physical transmission of bits over copper, fiber or wireless media.

KEY CHARACTERISTICS

- Safety-focused communication protocol
- Independent of underlying IP network
- Interoperable between devices from different manufacturers
- High availability through redundancy and supervision
- Meets railway standards and safety requirements

RaSTA DEPLOYMENT IN RAILWAY SYSTEMS

RaSTA enables safe, reliable and interoperable communication between safety-critical railway devices.

RaSTA
Safe Communication for Railways

RAILWAY DEVICES (EXAMPLES)

Control Centre (SCADA)
Central monitoring and operation of the network

Electronic Interlocking
Ensures safe routing of trains and interlocking logic

Signal Controller
Controls railway signals along the track

Level Crossing Controller
Manages level crossing safety functions

Train Control System (ETCS)
Onboard and trackside train control and supervision

RaSTA

Safe Communication over IP Networks

- Redundant Communication
- Deterministic Behavior
- Fault Detection
- Safe State Handling

TYPICAL HARDWARE / SOFTWARE PLATFORM

INDUSTRIAL CPU
- ARM Cortex-A / Cortex-M
- Intel x86
- PowerPC

REAL-TIME OPERATING SYSTEM

RTOS
- Linux RT
- QNX
- VxWorks

COMMUNICATION INTERFACES
- Ethernet (10/100/1000BASE-T)
- UDP/IP Protocol
- Fiber Optic / Redundant Network

RUNS
- RaSTA Protocol Stack
- Railway Safety Application

RaSTA is a software protocol running on standard industrial hardware platforms.
It provides safe, reliable and interoperable communication for mission-critical railway applications.

INSIDE A RaSTA COMMUNICATION NODE

RaSTA is implemented as a set of software modules that work together to provide safe, reliable and deterministic communication.

RaSTA
Safe Communication for Railways

KEY RESPONSIBILITIES

SAFETY FIRST
Ensures that communication failures do not lead to unsafe behavior.

CONTINUOUS MONITORING
Every message is supervised for correctness, timing and integrity.

FAULT HANDLING
Detects faults, tries safe recovery and switches to safe state if needed.

DETERMINISTIC BEHAVIOR
Guarantees predictable and timely communication.

INTEROPERABLE
Works with other RaSTA nodes across standard IP networks.

RAILWAY APPLICATION
e.g., Signal Control, Interlocking, ETCS

RaSTA API
Service Access Point for Applications

RaSTA PROTOCOL STACK

CONNECTION MANAGER
Establishes, maintains and closes RaSTA connections.

STATE MACHINE
Manages protocol states (DOWN, CONNECTING, UP, DISCONNECTING).

SEQUENCE MANAGER
Generates, verifies and tracks sequence numbers. Detects missing, duplicate or out-of-order packets.

HEARTBEAT MANAGER
Sends and supervises heartbeat messages. Detects communication loss.

TIMER MANAGER
Measures message timing, delays and timeouts. Ensures messages arrive within allowed window.

RETRANSMISSION MANAGER
Requests and retransmits missing packets. Ensures complete and reliable delivery.

INTEGRITY CHECKER
Verifies security/integrity code (e.g., MD4). Rejects corrupted or tampered messages.

UDP SOCKET
Datagram Transport

ETHERNET DRIVER
Network Interface

NETWORK INTERFACE CARD (NIC)
Physical Layer Connectivity

HOW IT WORKS

1. Application creates a message (e.g., set signal to GREEN).
2. RaSTA stack adds safety information, checks state and prepares the message.
3. Message is sent over UDP/IP through the network.
4. Receiver runs the same checks (sequence, timing, integrity).
5. If everything is correct, the application receives the data.

All modules work together to guarantee functional safety:
Detect faults → Try safe recovery → If not possible → Move system to safe state.

TYPICAL HARDWARE PLATFORM
- Industrial CPU: ARM Cortex-A / Intel x86 / PowerPC
- Real-Time OS: Linux RT / QNX / VxWorks
- Interfaces: Ethernet, Fiber Optic, Redundant Network

HOW RaSTA WORKS | FIRST 3 STEPS

RaSTA
Safe Communication for Railways

Establishing a **safe connection** between two RaSTA communication partners.

RaSTA connections are connection-oriented. Before any data is exchanged, the partners establish a **supervised and safe communication channel.**

1 CONNECTION REQUEST (CR)

Initiate connection

Partner A sends a Connection Request (CR) to Partner B.

Contains:

- Protocol version
- Sender identity
- Supported options
- Initial sequence number
- Integrity protection (MD4)

Purpose: Ask to establish a RaSTA connection and propose communication parameters.

2 CONNECTION RESPONSE (CC)

Accept or reject

Partner B evaluates the request and sends a Connection Response (CC).

Contains:

- Accept / Reject status
- Selected options
- Initial sequence number
- Integrity protection (MD4)

Purpose: Confirm or reject the connection and agree on communication parameters.

3 CONNECTION CONFIRM (CN)

Finalize connection

Partner A confirms the response with a Connection Confirm (CN).

Contains:

- Confirmation of parameters
- Initial sequence number
- Integrity protection (MD4)

Purpose: Finalize the connection. The connection is now established and supervised.

After Step 3, the connection is in the UP (UPPER) state.

Heartbeats are exchanged to supervise the connection and ensure both partners are alive.

HOW RaSTA WORKS | NEXT 3 STEPS

RaSTA
Safe Communication for Railways

Reliable data exchange, continuous supervision, and safe termination.

After the connection is established, **data is exchanged safely.**
The connection is **continuously supervised** using heartbeats.
If communication can no longer be guaranteed, the connection is **closed safely.**

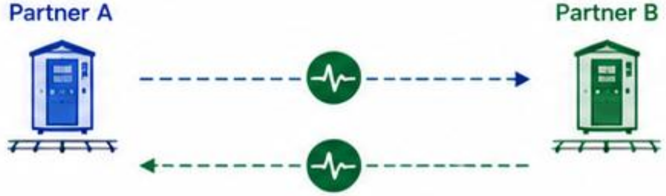
4 DATA TRANSFER
Exchange messages safely

Application data is sent with sequence numbers.
The receiver acknowledges each message.

Key Features:
- Sequence number supervision
- Integrity protection (MD4)
- Duplicate and out-of-order detection
- Retransmission if required

Purpose: Ensure reliable and ordered delivery of safety-critical data.

5 HEARTBEAT SUPERVISION
Keep connection alive

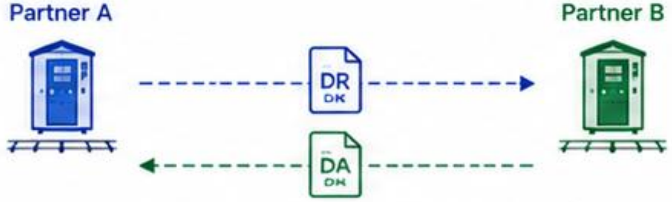


Both partners send heartbeats at regular intervals.
If a heartbeat is missed, a fault is assumed.

Key Features:
- Adaptive heartbeat timing
- Missed heartbeat detection
- Triggers safety actions if supervision fails

Purpose: Detect communication failures early and maintain connection supervision.

6 DISCONNECTION
Close connection safely



One partner initiates a Disconnect Request (DR).
The other confirms with a Disconnect Acknowledge (DA).

Key Features:
- Controlled connection termination
- Confirmation ensures both sides close safely
- Resources are released

Purpose: Terminate the connection in a controlled and safe manner.

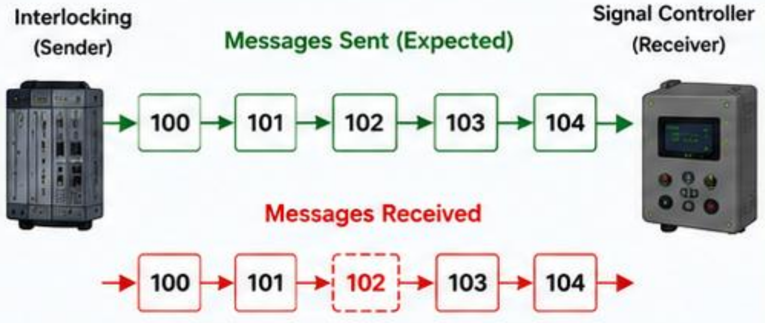
RaSTA ensures that communication is reliable, supervised, and safe from connection establishment to disconnection.

KEY SAFETY MECHANISMS – FIRST 3

1 SEQUENCE NUMBERS (SN)

Detects lost, duplicate, or out-of-order messages.

Example:



Message 102 is missing.
Retransmission requested.

Purpose: Ensures that all messages arrive in the correct order without loss or duplication.

2 INTEGRITY CHECKING (SECURITY CODE)

Detects corrupted or modified messages.

Example:

Original Message Sent

Command:
Set Signal = GREEN

SN: 101

Security Code:
A7 3F 9C 21

Received Message

Command:
Set Signal = GREEN

SN: 101

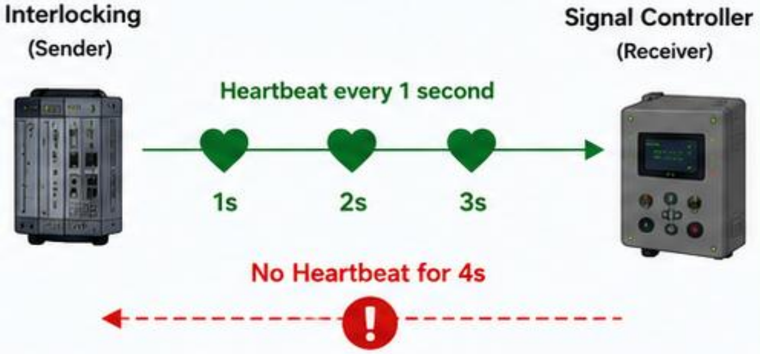
Security Code:
5B 2D 7E 10

Security code mismatch.
Message rejected.

Purpose: Guarantees message integrity by verifying that data is not corrupted or altered.

3 HEARTBEATS (HB)

Ensures both devices are still communicating.



Heartbeat timeout.
Connection closed.
System moves to safe state.

Purpose: Monitors the connection health and detects failures quickly.

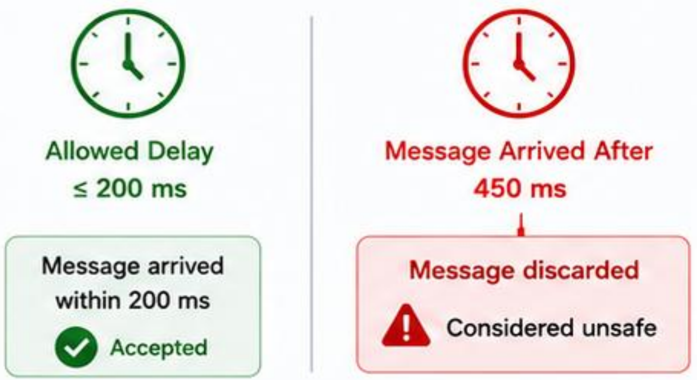
These mechanisms work together to detect faults, ensure reliable communication, and move the railway system to a safe state.

KEY SAFETY MECHANISMS – NEXT 3

4 ADAPTIVE TIMING

Detects messages that arrive too late or too early.

Example:

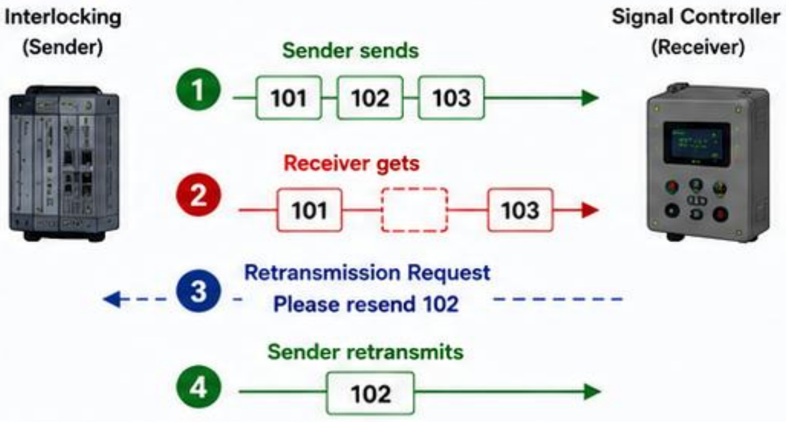


Purpose: Prevents outdated or premature messages from affecting system behavior.

5 RETRANSMISSION

Recovers lost messages by requesting and resending them.

Example:



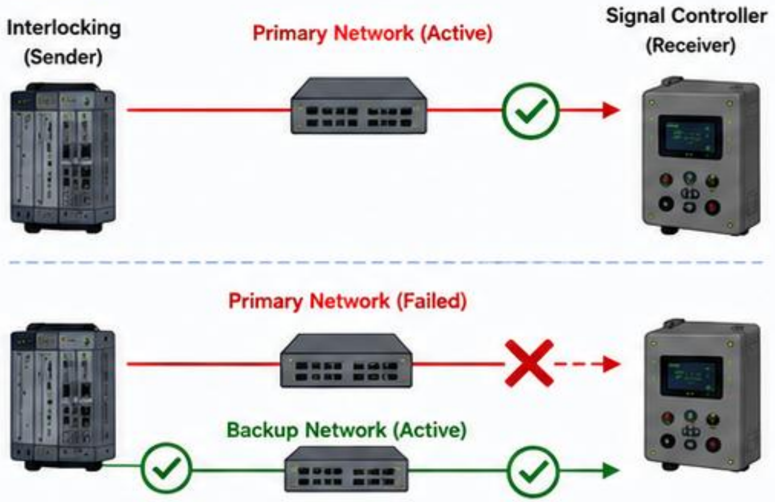
Result: Missing messages are recovered and the communication remains complete and reliable.

Purpose: Ensures no message is lost and the sequence is complete.

6 REDUNDANCY

Maintains communication if one network channel fails.

Example:



Purpose: Provides continuous communication by switching to a backup channel when the primary channel fails.

These mechanisms ensure reliable, complete, and timely communication — keeping railway operations safe and dependable.

OPERATIONAL EXAMPLE

Lost Packet Detection and Retransmission in RaSTA

RaSTA
Safe Communication for Railways

RaSTA continuously supervises sequence numbers, timestamps and integrity.
If a message is lost or faulty, it is detected and recovered—otherwise the connection is safely closed.

Sender (Interlocking)

Receiver (Signal Controller)

Step	Action	Message / Sequence No.	Result
1	Send Packet 101	Packet 101 (Seq: 101) →	Received (Seq: 101) ✓
2	Send Packet 102	Packet 102 (Seq: 102) →	Received (Seq: 102) ✓
3	Send Packet 103 (102 is lost)	Packet 103 (Seq: 103) → ✗ LOST	Not received ✗
4	Detect missing 102		Missing detected !
5	Request retransmission	← Retransmission Request (Seq: 102) →	Retransmission requested ✓
6	Retransmit Packet 102	Packet 102 (Seq: 102) →	Received (Seq: 102) ✓
7	Receive in correct order		✓ All received in order: 101, 102, 103 ✓
8	Continue communication	Heartbeat / Continue →	Communication continues ✓

What happens?

1. The sender transmits Packet 101.
2. The sender transmits Packet 102.
3. Packet 103 is sent, but Packet 102 is lost in the network.
4. The receiver detects that sequence 102 is missing.
5. The receiver sends a retransmission request for Packet 102.
6. The sender retransmits Packet 102.
7. The receiver now has 101, 102, 103 in the correct order.
8. Heartbeats continue and the connection remains in UP state.

Result

No unsafe action is triggered. Communication remains safe and reliable.

RaSTA ensures that communication errors are detected immediately and corrected.
If recovery is not possible, the connection is safely closed to protect the railway system.

LOST PACKET DETECTION & RETRANSMISSION

RaSTA ensures complete and ordered delivery of safety-critical messages.

RaSTA
Safe Communication for Railways

SCENARIO

The Interlocking (Sender) sends a command to the Signal Controller (Receiver):

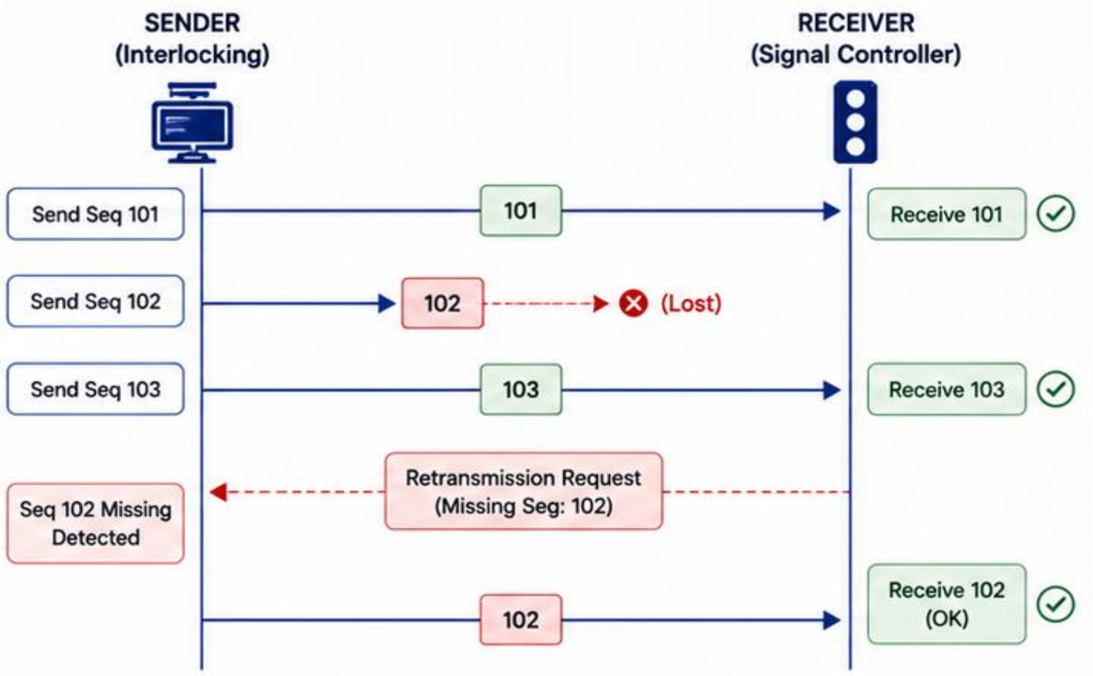
Command: "Set Signal 12 = GREEN"

PACKETS SENT BY SENDER

Seq: 101	Seq: 102	Seq: 103
Data	Data	Data
✓	✗	✓

✓ Delivered   ✗ Lost in network

WHAT HAPPENS ON THE NETWORK



All packets received in correct order: 101 → 102 → 103
Communication continues safely.

PROBLEM

Packet with Sequence Number 102 is lost during transmission. Receiver detects the missing sequence.

HOW RaSTA SOLVES IT

- Sequence Numbers detect missing packets.
- Receiver sends a Retransmission Request.
- Sender retransmits the missing packet.
- Communication continues safely.

TECHNOLOGY USED

- Sequence Numbering
- ACK / Retransmission Mechanism
- Timers (Retransmission Timeout)
- UDP over IP
- Integrity Code (MD4)

IF NOT SOLVED – WHAT MIGHT HAPPEN?

- Incomplete commands executed.
- Wrong signal aspect or missing safety command.
- Possible collisions or derailments.
- Risk to passengers, staff and infrastructure.

KEY TAKEAWAY

RaSTA detects any lost packet using sequence supervision and automatically recovers it through retransmission. This ensures that safety-critical information is always complete and reliable.

"RaSTA does not ignore failures. It detects them, recovers safely, or stops communication."

CORRUPTED MESSAGE DETECTION

RaSTA ensures the integrity of every message and rejects any data that has been altered.

RaSTA
Safe Communication for Railways

SCENARIO

The Interlocking sends a speed limit command to the Train Control System (ETCS).

Intended Message:
Speed Limit = 80 km/h

DURING TRANSMISSION

Due to noise or hardware fault, some bits are changed in the data.

Field	Intended Data	Corrupted Data
Speed Limit	80 km/h	180 km/h
Binary (Exemple)	0101 0000	1011 0100

- Original / Correct Data
- Corrupted Data

WHAT HAPPENS ON THE NETWORK

SENDER (Interlocking)

RECEIVER (Train Control System)

1. **Send Message**
   Speed Limit = 80 km/h
   (Seq: 145)

Data + Integrity Code (MD4)

Message Received
Speed Limit = 180 km/h
(Seq: 145)

2. **Integrity Check**
   Recalculate MD4 and compare

Reject / Discard
Integrity Code Mismatch!

3. **No ACK Received**
   Sender assumes message was not accepted

Retransmission Triggered

5. **Correct Message Received**
   Speed Limit = 80 km/h
   (Seg: 145)

Only messages with valid integrity are accepted.
Corrupted messages are always rejected.

PROBLEM

- Data is corrupted during transmission.
- Receiver cannot trust the content.
- Using wrong data can lead to dangerous decisions.

HOW RaSTA SOLVES IT

- Every message contains an Integrity Code (MD4).
- Receiver recalculates the code.
- If codes don't match, the message is rejected.
- Sender retransmits the message.

TECHNOLOGY USED

- Integrity Code (MD4)
- CRC/Hash based verification
- Sequence Numbers
- ACK / NACK Mechanism
- Timers for Retransmission

IF NOT SOLVED – WHAT MIGHT HAPPEN?

- Train receives wrong speed limit (e.g., 180 instead of 80).
- Overspeed may occur.
- Risk of derailment, collision or infrastructure damage.
- Threat to passenger safety and railway operations.

KEY TAKEAWAY

RaSTA protects message integrity using cryptographic verification.
Corrupted data is detected, rejected, and retransmitted – ensuring safe and reliable communication.

> *RaSTA does not deliver wrong data.*
> *It delivers correct data –*
> *or no data at all.*

OPERATIONAL EXAMPLE 3

DELAYED MESSAGE DETECTION

RaSTA ensures that time-critical messages arrive within an acceptable time window.
Messages that are delayed beyond the limit are rejected to prevent unsafe decisions.

RaSTA
Safe Communication for Railways

SCENARIO

An emergency braking command is sent from the Control Centre to the Train.

Intended Message:
EMERGENCY BRAKE = TRUE

TIMING REQUIREMENT

Max. allowable delay (Tmax):
500 ms

TIMELINE

Message Sent	t = 0 ms
Message Arrives	t = 900 ms
Delay	900 ms (> 500 ms)

Message arrives too late!
It is outside the valid time window.

WHAT HAPPENS ON THE NETWORK

SENDER (Control Centre / Interlocking) — **NETWORK** — **RECEIVER (Train / ETCS Onboard)**

1. **Send Message**
EMERGENCY BRAKE = TRUE
Timestamp: t = 0 ms

Network Delay
Caused by congestion / high load
Delay: 900 ms

2. **Message Arrives**
Timestamp in message: 0 ms
Arrival time: t = 900 ms

3. **Timing Check**
Current time – Timestamp
= 900 ms > 500 ms (Tmax)

4. **Message Rejected**
Discarded as outdated

Reject / Discard
No action taken based on this message

Only messages within the allowed time window are accepted.
Late messages are discarded to keep the system safe.

PROBLEM

- Due to network congestion, the message is delayed.
- Late information can no longer represent the real system state.
- Acting on old data can be dangerous.

HOW RaSTA SOLVES IT

- Every message has a Timestamp.
- Receiver checks the age of the message.
- If delay > Tmax, the message is rejected.
- System continues to operate with safe data.

TECHNOLOGY USED

- Timestamps in each message
- Adaptive Timing (Tmax)
- Timer Manager
- Message Age Verification
- UDP over IP

IF NOT SOLVED – WHAT MIGHT HAPPEN?

- Emergency brake may be applied too late.
- Train may not stop in time.
- Risk of collision, overspeed, or derailment.
- Threat to passenger safety and infrastructure.

KEY TAKEAWAY

RaSTA ensures messages are not only correct, but also timely.
Delayed messages are detected and rejected to guarantee safe and deterministic behaviour.

"In safety-critical railway systems, old information can be as dangerous as wrong information."

OPERATIONAL EXAMPLE 4

HEARTBEAT FAILURE

RaSTA continuously supervises the communication link using heartbeat messages.
If heartbeats stop, the connection is closed and the system moves to a safe state.

RaSTA
Safe Communication for Railways

SCENARIO

Two RaSTA nodes are connected between the Interlocking and a Field Device.

Heartbeat Interval: 1 second

WHAT IS A HEARTBEAT?

A small supervision message sent periodically to confirm that the communication partner is alive and the connection is healthy.

EXPECTED BEHAVIOR

- ✓ Heartbeat received every 1 second
- ✓ Connection remains active

IN THIS SCENARIO

- ✗ Heartbeats stop arriving (for 4 seconds)

WHAT HAPPENS ON THE NETWORK

NODE A (Interlocking) → NODE B (Field Device)

TIME	
t = 0 s	Heartbeat ✓ Received
t = 1 s	Heartbeat ✓ Received
t = 2 s	Heartbeat ✓ Received
t = 3 s	Heartbeat ✓ Received
t = 4 s	Heartbeat Missing...
t = 5 s	Still Missing...
t = 6 s+	Heartbeat Timeout! (× 3 missed heartbeats) → Connection Closed Safe State Activated

MONITORING WINDOW
Heartbeat Interval: 1 s
Timeout Threshold: 3 seconds
(3 missed heartbeats)

If heartbeats stop, RaSTA closes the connection and moves the system to a safe state.
This prevents uncontrolled or unsafe behavior.

PROBLEM

- Node B stops responding (crash, power loss, or network issue).
- No heartbeat messages are received.
- Connection health becomes unknown.

HOW RaSTA SOLVES IT

- Heartbeat messages are sent periodically.
- If 3 consecutive heartbeats are missed, a timeout occurs.
- Connection is closed automatically.
- System enters a defined safe state.

TECHNOLOGY USED

- Heertbeat Mechanism
- Timer Manager
- Connection State Machine
- Timeout Supervision
- UDP over IP

IF NOT SOLVED – WHAT MIGHT HAPPEN?

- System assumes communication is still alive.
- Old commands may continue to be valid.
- Unsafe movement authority may remain active.
- Risk of collision or derailment.
- Threat to passenger safety and infrastructure.

KEY TAKEAWAY

RaSTA's heartbeat supervision detects communication failures quickly.
When a failure is detected, the connection is closed and the system moves to a safe state, preventing unsafe actions caused by lost or failed communication.

“No heartbeat = No trust.
No trust = Safe state.”

OPERATIONAL EXAMPLE 5

PRIMARY NETWORK FAILURE

RaSTA maintains communication continuity by automatically switching to a redundant network when the primary network fails.

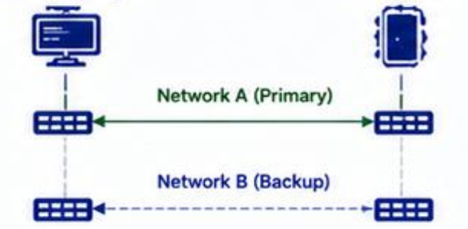
RaSTA
Safe Communication for Railways

SCENARIO

Two RaSTA nodes communicate using two independent Ethernet networks: Primary and Backup.

During operation, the primary network (Network A) fails.

NETWORK SETUP



NORMAL OPERATION

All data is transmitted over the Primary Network (A).

WHAT HAPPENS ON THE NETWORK

NODE A (Interlocking) | NETWORKS | NODE B (Field Device)

1. **Normal Operation** – Data is sent over Primary Network (A).
   Data / Heartbeat – Network A (Primary)
2. **Primary Network Failure** – Cable damage / Switch failure / Power loss on Network A.
   Link Down
3. **Failure Detection** – Loss of link detected by RaSTA (no heartbeat/ACK).
   Failure Detected
4. **Automatic Switch-Over** – RaSTA switches to Backup Network (B) within milliseconds.
   Switch to Backup Network (B) – Network B (Backup)
5. **Communication Restored** – Data transmission continues over Backup Network (B).
   Data / Heartbeat

RaSTA keeps the communication alive by using the redundant path. No manual intervention is required.

PROBLEM

- The primary network becomes unavailable.
- Communication would be interrupted without redundancy.
- Trains and safety systems could lose critical information.

HOW RaSTA SOLVES IT

- Continuous monitoring of both networks.
- Automatic failure detection.
- Instant switchover to the backup network.
- Communication resumes without data loss (sequence & retransmission handled by RaSTA).

TECHNOLOGY USED

- Redundant Ethernet Networks
- Link Monitoring
- Automatic Channel Switch-Over
- RaSTA Connection State Machine
- Sequence Numbers & Retransmission

IF NOT SOLVED – WHAT MIGHT HAPPEN?

- Loss of communication between critical devices.
- Interlocking may not receive field status.
- Wrong signal aspect or point position.
- Possible collisions, derailments, or service halt.

KEY TAKEAWAY

RaSTA's redundancy mechanism ensures high availability and fault tolerance.
If the primary network fails, RaSTA automatically uses the backup network to keep railway operations safe.

> *"Redundancy is not an option in railways – it is a requirement for safety and reliability."*

REAL-WORLD APPLICATIONS OF RaSTA

RaSTA ensures **safe** and **reliable** communication across modern railway systems.

RaSTA
Safe Communication for Railways

ELECTRONIC INTERLOCKINGS

- Controls signals, points and track routes
- Exchanges safety-critical commands and status
- Ensures safe route setting

ETCS / TRAIN CONTROL SYSTEMS

- Transmits movement authority and train data
- Provides continuous supervision
- Supports safe train separation

LEVEL CROSSINGS

- Coordinates barriers, lights and warning systems
- Exchanges real-time status information
- Prevents dangerous situations

CONTROL CENTRES

- Connects field devices with operations
- Centralized monitoring and diagnostics
- Enables faster and safer decision making

OTHER RAILWAY APPLICATIONS

- Trackside equipment communication
- Station systems
- Diagnostics and maintenance systems

RaSTA is used wherever safety-critical information must be exchanged reliably, deterministically and securely.

ADVANTAGES & LIMITATIONS OF RaSTA

RaSTA is designed to ensure functional safety in railway communication.

RaSTA
Safe Communication for Railways

ADVANTAGES

FUNCTIONAL SAFETY COMPLIANT
Designed according to DIN VDE V 0831-200 and DIN EN 50159 for safety-related communication.

HIGH COMMUNICATION SAFETY
Detects communication faults such as loss, delay, duplication, corruption and replay.

FAIL-SAFE OPERATION
If communication can no longer be trusted, RaSTA disconnects and supports a safe state.

HIGH AVAILABILITY
Redundancy layer allows communication to continue even if one channel fails.

INTEROPERABILITY
Standardized protocol allows equipment from different manufacturers to communicate.

DETERMINISTIC COMMUNICATION
Adaptive timing and supervision provide predictable and timely message delivery.

LIMITATIONS

MORE COMPLEX THAN TCP/IP
RaSTA adds multiple safety mechanisms, making the implementation more complex.

HIGHER COMMUNICATION OVERHEAD
Additional headers, sequence numbers and safety checks increase bandwidth usage.

REQUIRES SYNCHRONIZED IMPLEMENTATION
Correct operation depends on precise implementation and synchronization of both partners.

SPECIFIC TO RAILWAY APPLICATIONS
RaSTA is optimized for railway signalling and may not be suitable for general-purpose systems.

DOES NOT REPLACE APPLICATION-LEVEL SAFETY
Safety also depends on correct application logic and system design.

RaSTA prioritizes safety and reliability over pure performance, making it ideal for mission-critical railway systems.

SAFE | RELIABLE | INTEROPERABLE | RAILWAY-FOCUSED

CONCLUSION

RaSTA
Safe Communication for Railways

RaSTA provides **safe**, **reliable** and **deterministic** communication for safety-critical railway applications.

FUNCTIONAL SAFETY

RaSTA detects communication faults and prevents them from becoming safety hazards.

RELIABILITY & AVAILABILITY

Through sequence monitoring, adaptive timing, retransmission and redundancy, RaSTA ensures high availability.

SECURE & DETERMINISTIC

Integrity protection, replay prevention and deterministic behaviour guarantee trustworthy message exchange.

STANDARDIZED & INTEROPERABLE

Based on DIN VDE V 0831-200 and DIN EN 50159, RaSTA enables interoperability between devices from different manufacturers.

RAILWAY FOCUSED

Designed specifically for railway signalling and control systems, where safety is more important than performance.

**RaSTA does not prevent every communication failure—
it ensures that communication failures never become safety hazards.**

Thank you for your attention!
Questions and Discussion